

Resemblance Nominalism: A Solution to the Problem of Universals, by Gonzalo Rodriguez-Pereyra. Oxford: Clarendon Press, 2002. Pp. xii + 238. H/b: £35.00.

According to Resemblance Nominalists, facts about resemblance are primitive; all the other facts about the world are ultimately grounded in the facts about which particular things resemble one another. Gonzalo Rodriguez-Pereyra's new book performs a welcome service in resuscitating this important view, which has suffered a few decades of undeserved neglect. Rodriguez-Pereyra works out original accounts of the nature of the primitive resemblance facts, and the way in which they ground other kinds of facts. And he develops a detailed argument for the view: in the barest outline, we should believe it because it provides a solution to the "problem of universals", and is better than other putative solutions in respect of certain kinds of economy and explanatory power. But Rodriguez-Pereyra has much that is original and interesting to say about the nature of the problem, and about the factors relevant to deciding between the solutions.

Let's begin with the nature of the primitive resemblance facts. According to Rodriguez-Pereyra, these facts have the form 'x resembles y to degree n '. The objects that can play the role of n are not, as one might have expected, the reals between 0 and 1, but the positive integers. As Rodriguez-Pereyra recognises, this notion is rather different from anything we might ordinarily mean by 'resemblance'. Even its formal properties are unfamiliar. For Rodriguez-Pereyra there is no maximal degree of resemblance: although each object resembles itself at least as much as it resembles anything else, the degree of resemblance between two objects can be greater than the degree to which some third object resembles itself.

To introduce the intended interpretation of this unfamiliar primitive, Rodriguez-Pereyra provides the following gloss:

- (*) x resembles y to degree n iff exactly n of the classes having both x and y as members are *property classes*.

(‘Property class’ is Rodriguez-Pereyra’s term for what Lewis and Armstrong call ‘perfectly natural classes’.) Of course, (*) isn’t supposed to be an *analysis*. In fact, Rodriguez-Pereyra’s choice of the three-place notion of resemblance as primitive is largely motivated by the desire to be able to give an adequate analysis of ‘property class’, a central concern of Resemblance Nominalists since Carnap. Carnap’s initial suggestion was that “quality classes” could be defined as maximal classes any two of whose members resemble one another. Nelson Goodman objected that this analysis and its successors falsely entail that no quality class can be a subclass of another quality class (the companionship difficulty) and that the union of the intersections of any three quality classes must itself be a subclass of some quality class (the imperfect community difficulty). Although Rodriguez-Pereyra’s property classes, unlike Carnap’s quality classes, are not assumed to have any very direct connection to the phenomenal realm—so that Goodman’s intuitions about possible configurations of colours, sounds, and the like are not immediately relevant—Rodriguez-Pereyra holds that an acceptable analysis of ‘property class’ must avoid the analogues of Goodman’s difficulties. In fact, he thinks that an acceptable analysis should entail *nothing* about the distribution of property classes, with the sole exception of the principle that nothing is a member of infinitely many property-classes—which he takes to be necessary and a priori—and its logical and set-theoretic consequences.

Obviously, Rodriguez-Pereyra’s numerical resemblance-facts give him a much richer basis from which to reconstruct the facts about property classes. But if we had only the degrees of resemblance among *particulars* to work with in our analysis, there would still be no way to make all the distinctions between possibilities to which Rodriguez-Pereyra is committed. For example, we couldn’t allow that it is possible that the only property classes are $\{a, b\}$, $\{b, c\}$ and $\{a, c\}$, and also possible that the only property classes are $\{a, b, c\}$, $\{a\}$, $\{b\}$ and $\{c\}$, since, given (*), these two theories have exactly the same consequences as regards the degrees of resemblance among a , b and c . To explain distinctions like this one, Rodriguez-Pereyra appeals to primitive degrees of resemblance between certain mathematical objects, namely *hereditary unordered pairs* of particulars—i.e. unordered pairs of particulars, unordered pairs of unordered pairs of particulars, etc. (He

holds—on the basis of an argument I found quite unconvincing—that it would be unacceptable to appeal to primitive resemblance facts involving sets of cardinality larger than two.) For example, if the first of the above theories were true, $\{a, b\}$ would not resemble $\{b, c\}$ to any degree, whereas if the second were true, these unordered pairs would resemble to degree 1. The proposed additional constraint can be expressed in full generality as follows:

(**) If x and y are hereditary unordered pairs of the same order, x and y resemble to degree n iff exactly n property classes include all the particulars which are members of the transitive closures of x or y .

Rodriguez-Pereyra proves that (*) and (**) suffice, in conjunction with the principle that nothing is a member of infinitely many property classes, to entail a biconditional giving necessary and sufficient conditions for a class to be a property class, in terms of ‘resembles to degree n ’: he proposes this as a definition. Since (*) and (**) are obviously consistent with any logically consistent theory about the distribution of property-classes, this definition entails no more about the distribution of property-classes than is entailed by the principle that nothing is a member of infinitely many of them.

In this respect, Rodriguez-Pereyra’s definition seems to improve on Carnap’s; but there is a price, which he never acknowledges. The Carnapian definition straightforwardly *entails* (given set-theory and the symmetry of resemblance) that two things resemble iff there is some quality class that has both as members. Rodriguez-Pereyra’s definition, by contrast, comes nowhere near entailing (*) and (**). It couldn’t, because (*) and (**) have consequences which don’t contain the expression ‘property class’, and are not logical or set-theoretic truths. It’s not a truth of logic or of set theory that there is *any* family F of classes of particulars satisfying the condition that the degree of resemblance of any two particulars is equal to the number of members of F having both as members, let alone the more complicated condition corresponding to (**). Thus, if Rodriguez-Pereyra is right, the claim that there is such a family must be counted as a sort of basic axiom of metaphysics: a necessary and a priori truth that cannot be derived from logic and definitions. As far as I can see, there isn’t even much prospect of deriving it from any substantially simpler set of

axioms.

This seems like a strange and implausible result. Typically in philosophy, proposed analyses or reductive definitions are defended on the grounds that they explain certain analytic or necessary truths involving the analysandum by reducing them to logic, or at least to a simpler set of basic axioms. (Russell's logicism is a paradigm case.) Rodriguez-Pereyra's method is the exact reverse of this. His aim is to find a definition which is compatible with the claim that there are (almost) *no* interesting nonlogical analytic truths about the distribution of property classes; to achieve this, he is happy to accept a complicated basic axiom involving his favoured primitive. Even if the positing of such "brute necessities" makes sense, surely it must be counted as a serious failure of economy by comparison with the obvious alternative of taking 'property class' as primitive. I found it surprising that the book gives no consideration at all to this sort of economy, in view of its otherwise thorough treatment of methodological issues.

Rodriguez-Pereyra does consider the view that 'property class' is primitive: he argues that his own view is explanatorily superior, since it can explain what makes classes be property classes. But one could just as easily argue that the other view is explanatorily superior, on the grounds that it can explain what makes objects resemble to a certain degree. I can't see what 'explanatory superiority' could sensibly be taken to mean in this context if not superiority in being able to derive necessary or a priori truths from definitions; by this measure, Rodriguez-Pereyra's view does much worse than the alternative. His other objections to the view seem to me to be equally weak. For example, he objects that a primitive fact about a class couldn't be constitutive of a particular's having a property: but if this objection were good it would tell equally against his own invocation of primitive resemblances among unordered pairs.

Besides the rather abstruse question 'What it is to be a property class?', Rodriguez-Pereyra also sets himself the task of answering a wide range of other questions of the form 'What is it to be **F**?—those for which '**F**' "corresponds to a property", in a sense which requires at least that the class of **F** things be a property class. His final answer is something like this: to be **F** is to resemble, to some degree, each of the things that are (in fact) **F**, and to be such that those things form a

property class. He takes this to be equivalent to a claim about the constitution of the *fact* that *a* is **F**: this fact is a *conjunctive* fact, whose conjuncts include all the facts of the form ‘*a* resembles *x* to some degree’, for each *x* that is a member of the class of **F**s, together with the fact that this class is a property class. All this applies *mutatis mutandis* to relations, the only difference being that in the case of relations, the relevant primitive resemblances are those among ordered *n*-tuples.

This proposal faces a very obvious objection: since something could be white even if the Taj Mahal had never existed, it can’t be that to be white is, in part, to resemble the Taj Mahal. The only way I can see to resist this objection is to accept Lewis-style modal realism—which, rather startlingly, is exactly what Rodriguez-Pereyra does. It would certainly be interesting if it could be shown that only modal realists could consistently accept the view that facts about the properties of objects are ultimately grounded in facts about resemblances. But in fact, there are other possibilities. Instead of explaining what makes an object **F** in terms of its resemblances to *specific* particulars, we could identify the fact that an object is **F** with some *general* fact about its distinctive place in the overall pattern of resemblances. In view of the increasing popularity of Shoemaker’s view that properties have their causal or nomic relations essentially, the idea that the “purely structural” facts suffice to determine the facts about which properties are instantiated can hardly be dismissed out of hand. Unfortunately, Rodriguez-Pereyra never considers this sort of view, since it doesn’t count for him as a form of Resemblance Nominalism, or of any of the other theories on the list of “solutions to the problem of universals” which he inherits from Armstrong.

The influence of Armstrong is also manifest in Rodriguez-Pereyra’s conception of this alleged problem as arising from the need to identify entities to play the role of *truthmakers* for certain kinds of truths. For Rodriguez-Pereyra, what’s crucial is the requirement that *distinct* truthmakers be found for the truths that *a* is **F** and that *a* is **G**, whenever ‘**F**’ and ‘**G**’ correspond to distinct properties. I found Rodriguez-Pereyra’s treatment of truthmaking rather perplexing. On the one hand, he is happy to admit conjunctive facts and facts about resemblance. On the other hand, he maintains that conjunctions are jointly made true by the truthmakers of their conjuncts, and that truths about resemblance are jointly made true by the two resembling particulars. Given that the

fact that a is $\mathbf{F}$ turns out on Rodriguez-Pereyra's account to be a conjunctive fact, this seems to entail that the truth that a is $\mathbf{F}$ is jointly made true by all the $\mathbf{F}$ particulars, together with whatever makes it true that they form a property class; exactly the same things jointly make it true that anything else is $\mathbf{F}$. But whatever reason Rodriguez-Pereyra might have for holding that the truth that a is $\mathbf{F}$ must have different truthmakers from the truth that a is $\mathbf{G}$, one would think there should be at least as much reason to hold that the truth that a is $\mathbf{F}$ must have different truthmakers from the truth that b is $\mathbf{F}$, when $a \neq b$. There may be a well-motivated theory of truthmaking that is both general enough to leave room for a debate between Nominalists and Realists, but strong enough to impose a serious constraint on the shape of that debate. But if there is, Rodriguez-Pereyra has not yet succeeded in articulating it.

Despite its shortcomings, this book merits close attention from everyone who cares about the metaphysics of properties. In providing an example of how one might develop a worked-out nominalist conception of the basic structure of reality while taking seriously the problems to which universals are supposed to provide a solution, Rodriguez-Pereyra has brought much new energy and clarity to a debate in which these commodities have of late been in short supply.

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